

Comprehensive Question Bank: Unit 1 — Operating Systems: Structure and Services

Section A: Multiple-Choice Questions (MCQs)

Topic 1.1.1: Operating System as the Central System Controller

1. What is an operating system?
A) A physical component that stores data permanently
B) A special program that manages hardware, runs software, and provides a user interface
C) A hardware device that connects the CPU to the monitor
D) A type of application software used for word processing
2. Which of the following is the BEST analogy for the role of an operating system?
A) A student who completes homework assignments
B) A school principal who manages classrooms, schedules, and resources
C) A textbook that contains all necessary knowledge
D) A classroom teacher who delivers lessons
3. Which was widely considered the world's first operating system?
A) GM-NAA I/O
B) Windows 1.0
C) Unix
D) MS-DOS
4. In what year was GM-NAA I/O developed?
A) 1945
B) 1969
C) 1956
D) 1981
5. Which company developed GM-NAA I/O, the first operating system?
A) General Motors
B) IBM
C) Bell Laboratories
D) Microsoft

6. Before operating systems existed, how were programs loaded onto computers?
- A) Programs were automatically loaded by the CPU
 - B) Human operators manually loaded programs one at a time using punch cards**
 - C) Programs were stored permanently in the processor
 - D) Programs were downloaded from a central server
7. Which of the following is NOT a function of an operating system?
- A) Controlling hardware devices
 - B) Designing websites and web pages**
 - C) Providing a user interface
 - D) Running application software
8. Which of the following is an example of an operating system?
- A) Microsoft Word
 - B) Android**
 - C) Google Chrome
 - D) Adobe Photoshop
9. The operating system sits between which two components in a computer system?
- A) The keyboard and the monitor
 - B) The user and the hardware**
 - C) The CPU and the graphics card
 - D) The hard drive and the printer
10. Which operating system is primarily used on Apple MacBook computers?
- A) Windows 11
 - B) macOS**
 - C) Linux
 - D) Android
11. The operating system manages a conflict between two programs that need the same hardware resource. This is most similar to which role?
- A) A student taking an exam
 - B) A school principal resolving a classroom scheduling conflict**
 - C) A librarian issuing a book
 - D) A teacher grading papers
12. Which of the following operating systems is most commonly used on servers and scientific computers?
- A) iOS
 - B) Android
 - C) Linux**
 - D) macOS

13. What was the main problem that motivated the creation of the first operating system?
- A) Computers were too expensive to maintain
 - B) Computers spent too much time idle, waiting for human operators to manually load programs**
 - C) Computers could not perform mathematical calculations
 - D) Computers had no way to display output
14. Which of the following BEST describes what an OS does for a user?
- A) It increases the physical speed of the CPU
 - B) It allows people to use computers without needing to understand how hardware works**
 - C) It permanently stores the user's personal files
 - D) It connects the computer to the internet
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Topic 1.1.2: Role in User–Hardware Interaction

1. What role does the OS play between the user and the hardware?
- A) It stores user data permanently on the hard drive
 - B) It acts as a translator, converting human actions into binary instructions the hardware understands**
 - C) It speeds up the CPU when the user clicks a button
 - D) It displays graphics directly without any translation
2. What language does the CPU understand?
- A) English
 - B) Urdu
 - C) Binary (zeros and ones)**
 - D) Assembly code only
3. When you press the letter 'A' on a keyboard, what binary code does the OS translate it into?
- A) 10000010
 - B) 01000001**
 - C) 11000001
 - D) 00100001
4. How quickly does the OS typically translate a keyboard press into a character on screen?
- A) Less than one millisecond**
 - B) About one second
 - C) About 10 seconds
 - D) About one minute

5. What would happen if every app developer had to directly program hardware instead of using the OS?
- A) Programs would run faster
 - B) Every developer would need to learn the electrical language of every different hardware device**
 - C) The OS would become unnecessary
 - D) Programs would use less memory
6. The OS creates a "universal layer" that means:
- A) All programs must be written in the same programming language
 - B) Programs can talk to the OS instead of to specific hardware, so the same app can run on different machines**
 - C) All computers must have the same hardware specifications
 - D) Users must use the same operating system as other users on the network
7. What is the correct sequence when a user presses the letter 'B' on a keyboard?
- A) Keyboard sends signal → OS catches signal → OS translates to binary → sent to screen driver → screen displays B**
 - B) Screen displays B → OS translates → keyboard sends signal
 - C) CPU processes directly → screen displays B → OS records the action
 - D) OS translates → keyboard receives signal → CPU processes → screen displays
8. If applications could directly control hardware without the OS, which of the following risks would arise?
- A) The screen would display images faster
 - B) A malicious app could secretly record keystrokes without the user knowing**
 - C) Programs would use less battery power
 - D) The CPU would run cooler
9. The analogy for the OS as a translator is most like:
- A) A student learning a new subject
 - B) An interpreter who translates between someone speaking Urdu and someone speaking Mandarin**
 - C) A textbook that contains information
 - D) A clock that keeps track of time
10. Why is the OS's translation process important for software development?
- A) It forces all apps to use the same programming language
 - B) It makes apps run slower but more securely
 - C) It means developers write one program that works across many hardware types, rather than writing separate code for each device**
 - D) It allows the CPU to perform calculations faster
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Topic 1.1.3: Responsibilities in Multi-User Environments

1. In a multi-user environment, what does the OS use to ensure users can only access their own files?
 - A) Physical locks on the computer
 - B) User isolation — each user's data is kept in a separate, protected space**
 - C) Different computers for each user
 - D) Manual file sorting by an administrator
2. Which of the following is NOT a responsibility of the OS in a multi-user environment?
 - A) Sharing resources fairly between users
 - B) Keeping each user's files private
 - C) Teaching users how to type faster**
 - D) Protecting shared resources from unauthorised access
3. In a school computer lab with 30 students on the same server, what prevents one student from accessing another student's files?
 - A) The OS enforces user isolation — each account has its own private space**
 - B) The network cable physically blocks access
 - C) Each student uses a different hard drive
 - D) The teacher monitors all activity manually
4. The OS keeps records of who logged in, when, and what actions were performed. This OS responsibility is called:
 - A) Resource sharing
 - B) User isolation
 - C) Accountability**
 - D) Protection
5. If one user's program crashes in a properly designed multi-user OS, what happens to other users' programs?
 - A) Nothing — other users' programs continue running normally because the OS isolates processes**
 - B) All programs on the system crash immediately
 - C) The server restarts and all work is lost
 - D) The OS shuts down to prevent further damage
6. Which term describes the OS fairly dividing the CPU, memory, and printer between multiple users?
 - A) User isolation
 - B) Accountability
 - C) Resource sharing**
 - D) Authentication

7. In a multi-user system, a user must provide a username and password before gaining access. This process is called:
- A) Resource sharing
 - B) User isolation
 - C) Authentication**
 - D) Scheduling
8. Why is multi-user OS management essential in online banking systems?
- A) It speeds up transaction processing
 - B) It allows the bank to save electricity
 - C) It ensures each customer's account information stays private and inaccessible to others**
 - D) It prevents the server from needing upgrades
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Topic 1.1.4: Creating and Managing User Accounts

1. Which account type has full control over a computer, including installing software and creating other accounts?
- A) Guest
 - B) Standard User
 - C) Administrator**
 - D) Viewer
2. A Standard User account is best described as one that:
- A) Has no access to any files on the computer
 - B) Can use installed programs and change personal settings, but cannot install new software or change system-wide settings**
 - C) Has full control equivalent to an Administrator
 - D) Can only browse the internet
3. Which account type gives temporary, very limited access to a computer with no ability to save personal files?
- A) Administrator
 - B) Standard User
 - C) Guest**
 - D) Developer
4. A student tries to install a game on a school computer using a Standard User account. What happens?
- A) The OS prevents the installation because Standard Users cannot install software**

- B) The game installs but only for that one student
 - C) The OS asks the student to pay for the game first
 - D) The game installs but the teacher is notified
5. In Windows, where do you navigate to create a new user account?
- A) Control Panel → System → Devices
 - B) Settings → Accounts → Family & other users → Add account**
 - C) Task Manager → New User → Add Account
 - D) Device Manager → Users → Add New
6. After creating a new Standard User account on a computer, what does that account have access to?
- A) Its own separate desktop, files, and settings — but not other users' files**
 - B) All files belonging to the Administrator
 - C) The ability to install any software
 - D) The ability to delete other user accounts
7. A Guest user saves a file to the desktop and then logs out. When they log back in, what happens to that file?
- A) The file is still there and accessible
 - B) The file is moved to a shared folder
 - C) The file is typically not saved permanently — Guest accounts do not retain personal files between sessions**
 - D) The file is sent to the Administrator's account
8. Which component of a user account uniquely identifies each person on the system?
- A) Username**
 - B) Password
 - C) Account type
 - D) Desktop wallpaper
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Topic 1.2.1: Kernel vs. Shell

1. What is the kernel?
- A) The outer part of the OS that receives user commands
 - B) The core part of the OS that directly controls hardware resources such as the CPU, memory, and devices**
 - C) A graphical interface for displaying icons and menus
 - D) A program that manages network connections only
2. What is the shell?
- A) The outer layer of the OS that interacts with the user, receives commands, and passes them to the kernel**

- B) The core of the OS that directly controls hardware
 - C) A special type of device driver
 - D) The physical casing of the computer
3. Which team at Bell Laboratories developed Unix in 1969, establishing the kernel/shell separation?
- A) Bill Gates and Steve Ballmer
 - B) Ken Thompson and Dennis Ritchie**
 - C) Alan Turing and John von Neumann
 - D) Linus Torvalds and Richard Stallman
4. In the car analogy for kernel and shell, what does the kernel represent?
- A) The engine hidden under the hood that does the actual work**
 - B) The steering wheel that the driver controls
 - C) The dashboard that displays information
 - D) The gear stick that changes speed
5. In the car analogy, what does the shell represent?
- A) The engine under the hood
 - B) The fuel tank
 - C) The steering wheel, dashboard, and pedals — the controls the driver interacts with**
 - D) The tyres that contact the road
6. Which of the following is an example of a graphical shell?
- A) Windows Command Prompt
 - B) Linux Terminal
 - C) Windows Desktop with icons and menus**
 - D) The kernel space of an operating system
7. Which of the following is an example of a command-line shell?
- A) macOS Finder
 - B) Windows Command Prompt**
 - C) Android home screen
 - D) Windows Desktop
8. What is the correct order of layers from user to hardware?
- A) Hardware → Kernel → Shell → User
 - B) User → Shell → Kernel → Hardware**
 - C) User → Kernel → Shell → Hardware
 - D) Shell → User → Hardware → Kernel

9. What does the shell do when it receives a command from the user?
- A) It executes the command directly on the hardware
 - B) It translates the command into a request and passes it to the kernel**
 - C) It stores the command in permanent memory
 - D) It sends the command to another user
10. Which operating systems are built on the kernel/shell separation originally established in Unix?
- A) Windows 95 and MS-DOS only
 - B) Linux, macOS, Android, and iOS**
 - C) Only Linux
 - D) Only macOS and iOS
11. In Windows, what command do you type in the Command Prompt to list files in the current directory?
- A) dir**
 - B) ls
 - C) list
 - D) show
12. What command in the Windows Command Prompt moves you to the Desktop folder?
- A) go Desktop
 - B) move Desktop
 - C) cd Desktop**
 - D) open Desktop
13. Which statement best describes why users never directly interact with the kernel?
- A) The kernel does not exist on most modern computers
 - B) The shell acts as the intermediary — users give commands to the shell, which passes them to the kernel**
 - C) The kernel is too slow to receive direct commands
 - D) Direct kernel access would make computers run faster but is unnecessary
14. The kernel handles which of the following tasks? (Select the BEST answer)
- A) Displaying wallpaper and icons on the desktop
 - B) Allowing users to type commands in a text window
 - C) Deciding which program runs on the CPU and for how long, managing memory allocation, and controlling devices**
 - D) Installing new applications from the internet
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Topic 1.2.2: OS Layers and Modular Design

1. How many layers are in the OS architecture described in this unit?
A) Three
B) Four
C) Five (Layers 0 through 4)
D) Seven
2. What does Layer 0 (the lowest layer) of the OS architecture contain?
A) Physical hardware — CPU, RAM, hard drive, GPU, network card
B) System libraries and APIs
C) The user interface and applications
D) Device drivers and hardware controllers
3. What is the role of Layer 1 (Hardware Abstraction Layer / HAL)?
A) To run user applications
B) To contain device drivers that translate OS instructions into hardware-specific language
C) To provide the graphical desktop interface
D) To manage network connections exclusively
4. Layer 2 of the OS architecture is the Kernel Layer. Which of the following is NOT a kernel layer function?
A) Process management
B) Memory management
C) Displaying the graphical user interface to the user
D) Handling system calls
5. Which layer (Layer 3) provides system libraries, APIs, and file system services?
A) Physical Hardware Layer
B) Hardware Abstraction Layer
C) Service / Library Layer
D) User Interface Layer
6. What is Layer 4 of the OS architecture?
A) The User Interface Layer — applications, desktop, touch interface
B) The Kernel Layer
C) The Physical Hardware Layer
D) The Device Driver Layer
7. In the OS layered model, each layer communicates with:
A) All other layers simultaneously
B) Only the layers directly above and below it
C) Only the layer directly above it
D) Only the physical hardware

8. Why is the layered OS design beneficial when a device driver has a bug?
- A) The entire OS must be reinstalled to fix it
 - B) Only the driver (Layer 1) needs to be updated — the higher layers do not need to change**
 - C) The bug automatically spreads to all layers and must be fixed everywhere
 - D) The OS prevents all programs from running until the bug is fixed
9. In the school analogy for OS layers, what do "guards and cleaners" represent?
- A) Teachers and students
 - B) The Hardware Abstraction Layer (device drivers and hardware controllers)**
 - C) The kernel layer (administration)
 - D) Physical hardware (building and desks)
10. In the school analogy, the school administration (schedules, rules, resource allocation) maps to which OS layer?
- A) Physical Hardware Layer
 - B) Hardware Abstraction Layer
 - C) Kernel Layer**
 - D) User Interface Layer
11. What is the main safety benefit of the OS layered design?
- A) It makes the OS run faster
 - B) Applications in the upper layers cannot directly access hardware — they must request it through the kernel, preventing accidents and attacks**
 - C) It reduces the number of device drivers needed
 - D) It allows users to install any software without administrator permission
12. The modular layered design of an OS is compared in the textbook to:
- A) A river flowing downhill
 - B) Building a skyscraper floor by floor, where each floor depends on the one below it**
 - C) A bicycle wheel with many spokes
 - D) A classroom with many students
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Topic 1.2.3: System Libraries and Device Drivers

1. What is a system library?
- A) A collection of ready-made, tested code that programs can use to perform common tasks without writing those tasks from scratch**
 - B) A physical storage room where hardware components are kept
 - C) A list of all installed applications on the computer
 - D) A special program that connects the OS to network servers

2. When a photo editing app needs to open a JPEG file, what does it use instead of writing its own image-decoding code?
- A) A device driver
 - B) The OS image-reading system library**
 - C) The kernel directly
 - D) The shell interface
3. Which of the following is an example of a task handled by a system library?
- A) Controlling the speed of the CPU fan
 - B) Rendering text fonts on screen so every app does not have to draw each letter pixel by pixel**
 - C) Managing the physical connections between the motherboard and storage drive
 - D) Setting the computer's clock time
4. What is a device driver?
- A) A person who installs hardware into computers
 - B) A special program that allows the OS to communicate with a specific hardware device**
 - C) A type of system library for network communication
 - D) The physical cable connecting a device to the computer
5. Which layer of the OS architecture do device drivers belong to?
- A) Layer 4 (User Interface Layer)
 - B) Layer 3 (Service/Library Layer)
 - C) Layer 2 (Kernel Layer)
 - D) Layer 1 (Hardware Abstraction Layer / HAL)**
6. Why does every hardware device need its own driver?
- A) Because each hardware device speaks a slightly different electrical language at the circuit level, requiring its own translator**
 - B) Because drivers contain the operating system's core code
 - C) Because drivers make hardware run faster
 - D) Because without drivers, hardware cannot be physically connected to the computer
7. Which of the following is an example of a device driver?
- A) Microsoft Word
 - B) NVIDIA GeForce graphics card driver**
 - C) The Windows shell
 - D) A system library for audio playback

8. What message indicates the OS is automatically finding and installing a device driver?
- A) "System update available — please restart"
 - B) "Windows has found new hardware and is installing drivers"**
 - C) "Low disk space — please delete files"
 - D) "Application not responding — please wait"
9. Where in Windows can you view the drivers installed for hardware devices?
- A) Task Manager → Processes
 - B) Device Manager**
 - C) Control Panel → Clock and Region
 - D) Settings → Privacy
10. In Device Manager, under which category would you find the driver for a graphics card?
- A) Sound, video and game controllers
 - B) Display adapters**
 - C) Network adapters
 - D) Disk drives
11. What is the analogy for system libraries in the textbook?
- A) A bus driver who transports passengers
 - B) Pre-prepared ingredients from a shop, rather than growing your own wheat and raising your own cow**
 - C) A translator who speaks multiple languages
 - D) A computer's hard drive that stores programs permanently
12. If you install the wrong device driver (for a different model than your hardware), what is most likely to happen?
- A) The driver works perfectly because all drivers are compatible
 - B) The hardware may not function correctly, behave unexpectedly, or cause system instability**
 - C) The OS automatically corrects the driver to match the hardware
 - D) The computer immediately displays an error and shuts down safely
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Topic 1.3.1: Process Life Cycle

1. What is a process?
- A) A physical component of the CPU
 - B) A program that is currently loaded into memory and being executed by the CPU**
 - C) A type of file stored on the hard drive
 - D) A hardware device connected to the computer

2. Who designed the operating system for the Apollo 11 spacecraft, which successfully managed a computer overload during landing?
- A) Bill Gates
 - B) Dennis Ritchie
 - C) Margaret Hamilton**
 - D) Alan Turing
3. During which stage of the process lifecycle does the OS load the program into memory and allocate resources?
- A) Creation (New)**
 - B) Execution (Running)
 - C) Termination
 - D) Waiting
4. A process is actively using the CPU to perform its tasks. Which state is it in?
- A) New
 - B) Ready
 - C) Running**
 - D) Terminated
5. A process has requested data from the internet and is now waiting for the response. Which state is it in?
- A) New
 - B) Running
 - C) Waiting (Blocked)**
 - D) Terminated
6. When a process finishes its task and the user closes the application, what does the OS do?
- A) Moves the process to the Ready state for future use
 - B) Terminates the process and releases all its memory and resources back to the system**
 - C) Saves the process permanently to the hard drive
 - D) Moves the process to Virtual Memory
7. In the Chrome browser process lifecycle, what happens during the "Creation" phase?
- A) The OS loads Chrome's code from storage into RAM and creates a new process with a unique Process ID**
 - B) Chrome displays a web page to the user
 - C) Chrome waits for the user to type a URL
 - D) The OS closes Chrome and frees memory
8. What triggers a process to move from the "Running" state to the "Waiting" state?
- A) The user types a command
 - B) The process requests an I/O operation (such as reading a file or waiting for a**

- network response) and must wait for it to complete**
- C) Another process with higher priority arrives
 - D) The process completes its task
9. What triggers a process to move from the "Waiting" state back to the "Ready" state?
- A) The user clicks on the application window
 - B) The I/O operation (file read, network response) that the process was waiting for completes**
 - C) The CPU becomes free
 - D) The OS automatically moves all waiting processes after 5 seconds
10. Which process state follows the "New" state when the OS admits the process into the system?
- A) Running
 - B) Ready**
 - C) Waiting
 - D) Terminated
11. In the process state diagram, which transition occurs when the CPU is allocated to a Ready process?
- A) Ready → Waiting
 - B) Ready → Running**
 - C) Running → Terminated
 - D) New → Terminated
12. What is the CORRECT order of the main process lifecycle stages?
- A) Execution → Creation → Termination
 - B) Creation → Execution → Termination**
 - C) Termination → Creation → Execution
 - D) Creation → Termination → Execution
13. In the Apollo 11 story, what did the OS do when the computer was overloaded during the lunar landing?
- A) It shut down all processes immediately
 - B) It ignored lower-priority tasks and ran all tasks equally
 - C) It automatically identified critical tasks (landing the spacecraft) and dropped lower-priority tasks to protect the mission**
 - D) It requested more memory from Mission Control
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Topic 1.3.2: Multitasking and Concurrency

1. What is multitasking in an operating system?
- A) Using two computers at the same time
 - B) The OS allowing more than one program to be open and usable at the same time,**

appearing to run simultaneously

- C) Having two CPUs working together
 - D) Running a single program that performs multiple functions
2. On a computer with a single CPU core, how does multitasking actually work?
- A) The CPU runs all programs simultaneously using parallel circuits
 - B) The CPU switches between processes so rapidly — thousands of times per second — that all programs appear to run at the same time**
 - C) The CPU pauses all programs and runs them one per minute
 - D) The OS stores all programs in RAM and the user switches between them manually
3. What is concurrency in an operating system?
- A) Multiple processes being active (started and not yet finished) at the same time, even if only one is actually using the CPU at any instant**
 - B) Running two CPUs in the same computer
 - C) The ability to copy files from one location to another simultaneously
 - D) The OS switching between two users who share a computer
4. Which analogy from the textbook best illustrates concurrency?
- A) A student studying one subject at a time
 - B) A master chef working on five dishes simultaneously, cycling between them so all finish at roughly the same time**
 - C) A single printer printing one page at a time
 - D) A train that carries passengers to one destination
5. In a time-slicing diagram, each process receives:
- A) All the CPU time it needs until completion before the next process starts
 - B) A tiny slice of CPU time in rapid rotation with other processes**
 - C) CPU time proportional to how long it has been waiting
 - D) CPU time only when the user clicks on that program's window
6. What is the key difference between multitasking and concurrency?
- A) Multitasking uses two CPUs; concurrency uses one
 - B) Multitasking describes multiple programs being open and usable; concurrency describes multiple processes being in-progress simultaneously (even if only one uses the CPU at a time)**
 - C) Multitasking is for programs; concurrency is for files
 - D) There is no difference — they mean the same thing
7. You are listening to music, typing a document, and downloading a file simultaneously. Which OS feature makes this possible?
- A) Virtual memory

B) Device drivers

C) Multitasking and concurrency

D) System calls

8. In the time-slicing diagram, which of the following sequences correctly represents how the CPU divides attention?

A) Music → Browser → Music → Chat App → Music → Browser (rapid rotation)

B) Music runs until finished, then Browser runs until finished

C) All programs run on the CPU simultaneously through the same circuit

D) The user manually selects which program gets CPU time each second

Topic 1.3.3: Process Scheduling Concepts

1. What is process scheduling?

A) The process of installing new software on a computer

B) The OS method for deciding which process gets CPU time next and for how long

C) A calendar system that shows when programs will be updated

D) The way files are organised on the hard drive

2. What does FCFS stand for?

A) Fast CPU First Served

B) First-Come, First-Served

C) File Check and File Swap

D) Full CPU Frequency Setting

3. In FCFS scheduling, which process gets the CPU first?

A) The process that needs the least time to complete

B) The process with the highest priority

C) The process that arrived first

D) The process that the user is currently looking at

4. In FCFS scheduling, when does a process start running?

A) Only after all processes that arrived before it have completely finished

B) As soon as it arrives, regardless of other processes

C) Based on how long it needs to run

D) Based on how much memory it requires

5. Given the FCFS example: P1 arrives at 0s (needs 5s), P2 arrives at 1s (needs 3s), P3 arrives at 2s (needs 2s). When does P2 start running?

A) At 1 second

B) At 3 seconds

C) At 5 seconds

D) At 7 seconds

6. Using the same FCFS example (P1: 0s/5s, P2: 1s/3s, P3: 2s/2s), when does P3 finish?
- A) At 4 seconds
 - B) At 7 seconds
 - C) At 10 seconds**
 - D) At 12 seconds
7. Using the same FCFS example, what is P3's waiting time?
- A) 2 seconds
 - B) 4 seconds
 - C) 6 seconds**
 - D) 8 seconds
8. What is the "convoy effect" in FCFS scheduling?
- A) When all processes finish at exactly the same time
 - B) When short processes are stuck waiting behind a long process, like fast cars stuck behind a slow lorry on a single-lane road**
 - C) When processes are reordered to optimise CPU efficiency
 - D) When the OS copies processes to virtual memory
9. Which of the following is an ADVANTAGE of FCFS scheduling?
- A) Short processes always run first
 - B) It is simple, easy to implement, and every process eventually gets served without being skipped**
 - C) It prevents the convoy effect
 - D) It is the most efficient scheduling algorithm for all situations
10. Which of the following is a DISADVANTAGE of FCFS scheduling?
- A) It is too complex to implement
 - B) It skips some processes entirely
 - C) Short processes may wait a very long time if they are queued behind longer processes**
 - D) It requires knowledge of how long each process will take before it arrives
11. In Windows, which tool allows you to see all currently running processes and their CPU usage in real time?
- A) Task Manager (opened with Ctrl + Shift + Esc)**
 - B) Device Manager
 - C) File Explorer
 - D) Control Panel
12. In the FCFS method, P1 arrives at 0s (needs 3s), P2 arrives at 1s (needs 5s), P3 arrives at 2s (needs 3s). What is P3's waiting time?
- A) 2 seconds

- B) 5 seconds
 - C) 6 seconds**
 - D) 8 seconds
13. FCFS scheduling is described as being similar to which real-world scenario?
- A) A hospital emergency room where the sickest patient is treated first
 - B) A queue at a shop counter where the first customer in line is served first**
 - C) A traffic light system that changes based on traffic density
 - D) A teacher calling on students based on who raises their hand
14. Which of the following describes why FCFS is NOT suitable for real-time operating systems?
- A) FCFS processes too many tasks simultaneously
 - B) FCFS cannot guarantee that critical tasks will be processed within strict time deadlines**
 - C) FCFS requires too much memory to run
 - D) FCFS only works with a single CPU
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Topic 1.4.1: Primary Memory (RAM)

1. What does RAM stand for?
 - A) Read and Access Memory
 - B) Random Access Memory**
 - C) Rapid Allocation Memory
 - D) Real-time Application Memory
2. What is RAM primarily used for?
 - A) Permanently storing files and documents
 - B) Temporarily holding programs and data that are currently active and in use by the CPU**
 - C) Storing the BIOS firmware
 - D) Managing network connections
3. Which of the following is a key characteristic of RAM?
 - A) It retains data permanently even when the computer is turned off
 - B) It is erased when the computer is turned off (volatile)**
 - C) It is slower than the hard drive
 - D) It stores programs that are not currently running
4. In the "desk and filing cabinet" analogy, what does the desk represent?
 - A) RAM — the fast, limited workspace where you keep what you are currently using**
 - B) The hard drive — large, permanent storage
 - C) The CPU — where calculations are performed
 - D) Virtual memory — overflow storage

5. The IBM 5150 (1981), the first popular personal computer, had how much RAM?
- A) 1 megabyte
 - B) 16 kilobytes**
 - C) 64 kilobytes
 - D) 1 gigabyte
6. Modern DDR5 RAM can transfer data at speeds exceeding:
- A) 500 MB per second
 - B) 7,000 MB per second
 - C) 50 GB per second**
 - D) 100 MB per second
7. Which of the following items is stored in RAM while a computer is running?
- A) Program files waiting to be installed
 - B) The operating system itself, all open applications, and data currently being processed**
 - C) Files that have been permanently deleted
 - D) Device drivers stored for future use
8. What happens to the data in RAM when the computer is shut down?
- A) It is automatically saved to the hard drive
 - B) It is moved to virtual memory for safekeeping
 - C) It is completely erased — RAM is volatile and does not retain data without power**
 - D) It is compressed and stored in a backup file
9. How does having more RAM improve computer performance?
- A) It makes the CPU run at a higher clock speed
 - B) More programs can be held in active memory at once, reducing the need to swap data to slower virtual memory**
 - C) It increases the speed of the internet connection
 - D) It decreases the amount of heat produced by the processor
10. Which tool in Windows shows a real-time graph of how much RAM is currently in use?
- A) Device Manager → Memory
 - B) Task Manager → Performance → Memory**
 - C) File Explorer → Storage
 - D) Settings → About → Memory Usage
-

Topic 1.4.2: Virtual Memory

1. What is virtual memory?
- A) A type of RAM that is faster than regular RAM
 - B) A technique where the OS uses part of the storage drive (HDD/SSD) as extra RAM**

when physical RAM is full

- C) Memory used exclusively for running the operating system
 - D) A type of memory built directly into the CPU
2. Virtual memory was invented at which university in 1962?
- A) MIT
 - B) Stanford University
 - C) University of Manchester (with the Atlas computer)**
 - D) Harvard University
3. Who led the team that invented virtual memory at the University of Manchester?
- A) Margaret Hamilton
 - B) Tom Kilburn**
 - C) Dennis Ritchie
 - D) Ken Thompson
4. In Windows, what is the storage area on the hard drive used for virtual memory called?
- A) Swap space
 - B) Page file**
 - C) Cache memory
 - D) Boot sector
5. In Linux/macOS, what is the storage area on the drive used for virtual memory called?
- A) Swap space**
 - B) Page file
 - C) Cache memory
 - D) Virtual drive
6. What is "paging" or "swapping" in the context of virtual memory?
- A) Moving process data between RAM and the storage drive as needed when RAM fills up**
 - B) Deleting old files from the hard drive to free up space
 - C) Copying files between two different hard drives
 - D) Partitioning the hard drive into separate sections
7. Approximately how fast is RAM (DDR5) compared to a regular SSD?
- A) About the same speed
 - B) About twice as fast
 - C) About 100 times faster (DDR5: ~50,000 MB/s vs SSD: ~500 MB/s)**
 - D) About 10 times faster
8. What is "thrashing" in the context of virtual memory?
- A) When the OS is so heavily using virtual memory (constantly swapping data in and out) that overall performance degrades significantly**

- B) When the hard drive physically fails
 - C) When two programs try to use the same memory address
 - D) When the CPU temperature exceeds safe limits
9. In the "desk and backpack" analogy for virtual memory, what does the backpack represent?
- A) RAM — the primary fast workspace
 - B) Virtual memory on the storage drive — overflow storage that is slower to access**
 - C) The CPU — the place where calculations happen
 - D) The operating system — the manager of all resources
10. What is the main trade-off of using virtual memory?
- A) Virtual memory increases the risk of data loss
 - B) Virtual memory allows more programs to run but at reduced speed, because storage drives are much slower than RAM**
 - C) Virtual memory increases the amount of physical RAM needed
 - D) Virtual memory prevents programs from accessing the CPU
11. In Windows, where do you go to view and change the virtual memory (paging file) settings?
- A) Settings → Storage → Virtual Memory
 - B) System Properties → Advanced → Performance Settings → Advanced → Virtual Memory → Change**
 - C) Task Manager → Memory → Page File
 - D) Device Manager → Memory → Paging
12. Which of the following storage types has the HIGHEST data transfer speed?
- A) RAM (DDR5) at ~50,000 MB/s**
 - B) NVMe SSD at ~7,000 MB/s
 - C) Regular SSD at ~500 MB/s
 - D) Hard Drive at ~100–150 MB/s
-

Topic 1.5: Processes vs. Threads

1. What is the key difference between a process and a thread?
- A) A process is faster than a thread; a thread is slower
 - B) A process has its own private memory space, while a thread shares memory with other threads within the same process**
 - C) A process is part of a thread; a thread contains many processes
 - D) A process runs on the GPU; a thread runs on the CPU

2. In the restaurant kitchen analogy, what does the entire kitchen represent?
- A) A process**
 - B) A thread
 - C) Virtual memory
 - D) The kernel
3. In the restaurant kitchen analogy, what do the individual cooks represent?
- A) Processes
 - B) Threads**
 - C) Device drivers
 - D) System calls
4. Which of the following statements about threads is TRUE?
- A) Threads within the same process share memory, making data sharing between them easy**
 - B) Threads have their own private memory space, completely isolated from other threads
 - C) Creating a new thread is more expensive than creating a new process
 - D) A thread crash cannot affect other threads in the same process
5. Which of the following is an example of two PROCESSES (not threads)?
- A) Two tabs inside the same Chrome window
 - B) Chrome browser and Microsoft Word running simultaneously**
 - C) The spell checker and the auto-save function in Microsoft Word
 - D) The video player and the download manager within one browser window
6. In the web browser example, what does the Chrome browser application as a whole represent?
- A) A process**
 - B) A thread
 - C) A system call
 - D) A device driver
7. Within a Chrome browser process, Thread 1 loads a webpage, Thread 2 plays a YouTube video, and Thread 3 downloads a file. These threads all share:
- A) The browser's memory, network connections, and settings**
 - B) Separate memory spaces and separate CPU time
 - C) Separate processes but with linked data
 - D) The same CPU core exclusively without switching
8. If a process crashes, what happens to other processes running on the same computer?
- A) All other processes also crash immediately
 - B) Other processes are typically unaffected because processes are isolated from one**

another

- C) Other processes pause temporarily until the crashed process restarts
 - D) Other processes move to virtual memory until the system stabilises
9. Which of the following statements correctly compares the resource cost of processes vs. threads?
- A) Processes are cheaper to create than threads
 - B) Threads are cheaper to create than processes because threads share the parent process's memory rather than requiring their own allocation**
 - C) Both processes and threads cost the same to create
 - D) Threads are more expensive because they require additional memory allocation
10. In Scenario A, you open 10 browser tabs in one Chrome window. In Scenario B, you open 10 separate Chrome windows. Which uses more RAM?
- A) Scenario A — because tabs share memory less efficiently than windows
 - B) Scenario B — because each window may be a separate process with its own memory allocation**
 - C) Both use exactly the same amount of RAM
 - D) Neither uses RAM because browsers use only virtual memory
-

Topic 1.5.1 & 1.5.2: Multithreading and Its Benefits

1. What is multithreading?
- A) The ability of a single process to run multiple threads simultaneously, allowing it to perform multiple tasks at once**
 - B) Running multiple processes on two separate CPUs
 - C) The OS switching between multiple applications every second
 - D) Using RAM and virtual memory at the same time
2. In a word processor that uses multithreading, which of the following can happen simultaneously?
- A) Only one task can happen at a time
 - B) Thread 1 accepts keyboard input while Thread 2 checks spelling and Thread 3 auto-saves**
 - C) Spelling check blocks typing until it finishes
 - D) Auto-save must complete before the user can type again
3. Which benefit of multithreading refers to applications staying responsive during heavy background work?
- A) Enhanced Performance

B) Improved Responsiveness

- C) Efficient Use of Resources
- D) Support for Concurrent Operations

4. A video game uses separate threads for graphics rendering, physics, sound, AI, and networking. Which benefit of multithreading does this illustrate?

A) Enhanced Performance — tasks divided into parallel threads complete more quickly

- B) Improved Responsiveness — the UI stays interactive
- C) Efficient Use of Resources — threads share memory
- D) Support for Concurrent Operations — multiple operations overlap

5. Why do threads use fewer system resources than creating a new process?

- A) Threads do not use any memory at all

B) Threads share the memory space and resources of their parent process, so the OS does not need to allocate a completely separate memory region

- C) Threads run on a different processor that requires less power
- D) Threads are stored in virtual memory, not RAM

6. Which of the following applications MOST benefits from multithreading?

- A) A simple clock that only displays the time
- B) A basic text file with no interactive features

C) A video editing application that plays back a preview while exporting a finished video and loading the next clip simultaneously

- D) A static image displayed on screen

7. The benefit of multithreading that allows a game to render graphics, calculate physics, and play sound at the same time is called:

- A) Improved Responsiveness
- B) Enhanced Performance

C) Support for Concurrent (parallel) Operations

- D) Efficient Use of Resources

8. In the multithreading sequential vs. parallel diagram, which sequence shows tasks done WITHOUT multithreading?

A) Type text → Check spelling → Save file (one task finished before the next begins)

- B) Type text + Check spelling + Save file (all simultaneously)
 - C) Save file → Type text → Check spelling
 - D) Check spelling (while typing) → Save file (in background)
-

Topic 1.6 & 1.6.1: System Calls and Their Types

1. What is a system call?
 - A) A phone call made to technical support when a computer fails
 - B) A formal request made by a program to the operating system to perform a hardware-level task that the program cannot do directly**
 - C) A function called between two different running programs
 - D) A command typed by a user into the command-line shell
2. System calls act as a bridge between:
 - A) Two different operating systems
 - B) User programs and the OS kernel**
 - C) The CPU and RAM
 - D) Two separate computers on a network
3. Why do programs need system calls instead of directly accessing hardware?
 - A) For security and stability — the OS validates every request before acting, preventing programs from corrupting hardware, memory, or other programs' data**
 - B) Because hardware is too fast for programs to communicate with directly
 - C) Because programs cannot store data in RAM without system calls
 - D) Because system calls make programs run faster
4. What happens when a program makes a system call?
 - A) The program crashes and must be restarted
 - B) The request crosses from user space into kernel space, where the OS validates and fulfills the hardware action, then returns the result**
 - C) The program pauses permanently until a human approves the request
 - D) The program copies itself to virtual memory
5. What does the `open()` system call do?
 - A) Opens a file for reading or writing by locating it on the storage drive and returning a file handle to the program**
 - B) Opens a new program window on the desktop
 - C) Starts a new thread within the current process
 - D) Opens a network connection to a remote server
6. What does the `read()` system call do?
 - A) Reads the current system time
 - B) Retrieves data from an open file or input device and passes it to the program**
 - C) Reads a list of all running processes
 - D) Reads the amount of available RAM

7. What does the `write()` system call do?
- A) Writes a program into memory for execution
 - B) Sends data from a program to a file on disk or an output device such as a printer or speaker**
 - C) Writes a new user account to the system
 - D) Writes a log entry to the OS event journal only
8. What does the `fork()` system call do?
- A) Divides a file into two equal parts
 - B) Creates a new process by duplicating the calling process**
 - C) Splits a thread into two separate threads
 - D) Copies a file from one folder to another
9. When you save a document in a word processor, which system call is primarily used?
- A) `open()`
 - B) `read()`
 - C) `write()`**
 - D) `fork()`
10. When you open a music file in a music player, which system call is FIRST used?
- A) `open()`**
 - B) `read()`
 - C) `write()`
 - D) `fork()`
11. After a music file is opened, which system call retrieves the audio data for playback?
- A) `open()`
 - B) `read()`**
 - C) `write()`
 - D) `fork()`
12. Approximately how many system calls does Windows have?
- A) About 50
 - B) About 100
 - C) Nearly 2,000**
 - D) About 500
13. The boundary in the system call diagram between "user space" and "kernel space" exists to:
- A) Prevent user programs from directly accessing hardware, ensuring the kernel validates all hardware requests for security and stability**
 - B) Speed up communication between the CPU and RAM
 - C) Separate the shell from the user interface
 - D) Allow two programs to share the same memory address
-

Topic 1.7.1: Files

1. What is a file?
 - A) A container that holds multiple folders
 - B) A collection of related data stored on a computer**
 - C) The OS method for organising data on a hard drive
 - D) A pointer that shows where data is located on disk
 2. Which of the following is NOT a type of data a file can contain?
 - A) Text (a story or essay)
 - B) Images (photos)
 - C) Audio (music)
 - D) Other complete file systems**
 3. A file named `homework.docx` — what does the `.docx` part tell the OS?
 - A) The file's size
 - B) The file's type — that this is a Microsoft Word document**
 - C) The file's creation date
 - D) The file's location on the drive
 4. Which of the following correctly lists the components that every file has?
 - A) Name, extension, location, and content**
 - B) Name, size, and password
 - C) Extension, author, and creation date only
 - D) Location, content, and driver
 5. Which file extension indicates an executable program?
 - A) .docx
 - B) .mp3
 - C) .exe**
 - D) .jpg
-

Topic 1.7.2: Folders

1. What is a folder (directory)?
 - A) A collection of related data stored on a computer
 - B) A container used to organise files and other folders in a logical, hierarchical structure**
 - C) The metadata attached to a file
 - D) The program used to open and view files

2. What type of structure do folders create when organising files?
 - A) A flat list where all files are at the same level
 - B) A hierarchical tree structure, with folders containing files and other folders**
 - C) A random order that the OS manages internally
 - D) A circular structure where folders link back to the root
 3. In the file path `C:\Users\Ahmed\Documents\Homework.docx`, what is `C:\` called?
 - A) The root folder — the top level of the file system hierarchy**
 - B) The file name
 - C) The file extension
 - D) The user's home folder
 4. Every file has a unique address that shows its location in the folder hierarchy. This is called a:
 - A) File extension
 - B) Metadata tag
 - C) File path**
 - D) Directory tree
 5. According to the textbook file system hierarchy, which folder contains both the "Ahmed" and "Sara" user folders?
 - *A) Users***
 - B) Program Files\
 - C) Documents\
 - D) Desktop\
-

Topic 1.7.3: Metadata

1. What is metadata?
 - A) The actual content stored inside a file (the text, images, audio)
 - B) Data about data — information that describes a file (name, size, dates, permissions) without being its actual content**
 - C) The programming code that runs when a file is opened
 - D) The backup copy of a file stored in virtual memory
2. In the book analogy for metadata, the TEXT INSIDE the book represents the file's:
 - A) Content**
 - B) Metadata
 - C) Extension
 - D) Path
3. In the book analogy, the COVER INFORMATION (title, author, publisher, year, page count) represents:
 - A) Content

B) Metadata

C) File extension

D) Directory

4. Which of the following is an example of file metadata?

A) The paragraphs of text in a Word document

B) The pixels of an image in a photo

C) The date a file was last modified and its file size

D) The music notes in an audio file

5. For a photo taken on a smartphone, which of the following is metadata (NOT the image itself)?

A) The colours in the sunset captured in the photo

B) The faces of people visible in the photo

C) The GPS coordinates of where the photo was taken

D) The brightness of the sky in the photo

6. How does the OS use metadata to help users?

A) The OS uses metadata to quickly find, sort, and display file information (like size, date, type) without opening the files themselves

B) Metadata allows the OS to increase the size of files automatically

C) Metadata stores backup copies of all files in RAM

D) The OS uses metadata to encrypt file contents

7. What security concern arises from photos containing GPS location metadata?

A) Photos become too large to send over the internet

B) A photo shared online could reveal the exact location where it was taken, potentially compromising the photographer's privacy or safety

C) GPS metadata causes photos to display incorrectly on some devices

D) GPS metadata makes photos incompatible with some operating systems

8. In Windows, where do you right-click a file to view its metadata (size, dates, type)?

A) Open With → Metadata

B) Right-click → Properties

C) Right-click → Format

D) Open → File → Info

Topic 1.7.4: File Systems

1. What is a file system?

A) A physical component inside the hard drive

B) The method the OS uses to organise, store, and retrieve files on a storage device

C) The folder hierarchy visible in File Explorer

D) A type of RAM used to temporarily store file data

2. Which file system is commonly used on USB flash drives due to its high compatibility with many devices?
- A) FAT32**
 - B) NTFS
 - C) EXT4
 - D) APFS
3. Which file system is used by Windows computers and offers large file support, detailed permissions, and journaling?
- A) FAT32
 - B) NTFS**
 - C) EXT4
 - D) APFS/HFS+
4. Which file system is commonly used by macOS and is optimised for SSDs with strong encryption?
- A) FAT32
 - B) NTFS
 - C) EXT4
 - D) APFS/HFS+**
5. Which file system is primarily used by Linux computers?
- A) FAT32
 - B) NTFS
 - C) EXT4**
 - D) APFS
6. What is a major limitation of the FAT32 file system?
- A) It is not compatible with any operating system
 - B) It cannot store folders, only files
 - C) It has a maximum file size limit of 4 GB per individual file**
 - D) It does not support file names longer than 8 characters
7. Which file system is compatible with BOTH Windows and Mac, and has no 4 GB file size limit, making it ideal for modern USB drives and SD cards?
- A) FAT32
 - B) NTFS
 - C) EXT4
 - D) exFAT**
8. What does "journaling" in a file system (like NTFS and EXT4) provide?
- A) Crash recovery — the file system logs changes before making them, so it can recover to a consistent state if the system crashes mid-write**

- B) A daily log of all user actions kept for accountability
 - C) A feature that automatically backs up files to cloud storage
 - D) A diary of all software updates applied to the operating system
9. What does the file system decide about each file stored on a disk?
- A) Where on the physical disk sectors each file is placed, and it keeps a record of that location so the file can be found later**
 - B) The file's name, extension, and content
 - C) How much RAM the file will use when opened
 - D) Which user created the file and their password
-

Topic 1.8.1: Real-Time Operating System (RTOS)

1. What is a Real-Time Operating System (RTOS)?
- A) An OS that runs programs faster than other operating systems
 - B) An OS designed to process tasks and deliver results within strict, guaranteed time limits called deadlines**
 - C) An OS that runs only one program at a time in real time
 - D) An OS optimised for streaming live video and audio
2. In an RTOS, what makes a "late" response a failure?
- A) In safety-critical systems, timing is as important as correctness — a correct answer delivered after the deadline can cause physical harm or system failure**
 - B) Late responses cost extra processing power
 - C) RTOS cannot handle late responses and crashes
 - D) Late responses use more memory than on-time responses
3. Which of the following applications MOST requires an RTOS?
- A) A word processing application for writing school essays
 - B) A web browser for browsing social media
 - C) An anti-lock braking system (ABS) in a car that must respond within milliseconds during a skid**
 - D) A music streaming application
4. What is the key characteristic that distinguishes an RTOS from a general-purpose OS (like Windows)?
- A) An RTOS can run more applications simultaneously
 - B) An RTOS guarantees that tasks will be processed within strict time deadlines; a general-purpose OS does not guarantee response times**
 - C) An RTOS requires less physical hardware to operate
 - D) An RTOS supports more device drivers than a general-purpose OS

5. Which of the following would NOT typically use an RTOS?
- A) Air traffic control systems
 - B) Industrial robot arms
 - C) A home computer used for web browsing and word processing**
 - D) Heart-monitoring devices (pacemakers)
6. Why would it be dangerous to use a general-purpose OS like Windows in a pacemaker?
- A) Windows may pause for variable amounts of time for updates, notifications, or background tasks — in a pacemaker, any unpredictable delay could be life-threatening**
 - B) Windows uses too much power and would drain the pacemaker battery
 - C) Windows cannot run on the small chips used in medical devices
 - D) Windows does not support the file systems used by medical equipment
7. What did the RTOS on the Apollo 11 guidance computer do when it was overloaded?
- A) It shut down all processes and restarted
 - B) It automatically dropped lower-priority tasks to ensure critical tasks (landing the spacecraft) were completed on time**
 - C) It requested additional computing power from Mission Control
 - D) It paused all tasks and waited for the overload to resolve itself
-

Topic 1.8.2: Embedded Operating System (EOS)

1. What is an Embedded Operating System?
- A) An OS installed inside a general-purpose computer for security
 - B) A small, highly efficient OS built permanently into a specific device to perform a fixed, limited set of tasks**
 - C) An OS that runs inside a virtual machine on another OS
 - D) An OS for running multiple programs on a network server
2. Which type of memory often stores an Embedded OS permanently inside a device?
- A) RAM
 - B) Virtual memory
 - C) ROM (Read-Only Memory)**
 - D) Cache memory
3. Which of the following is an example of a device that uses an Embedded OS?
- A) A laptop running Windows
 - B) A washing machine that controls wash cycles and water levels**
 - C) A gaming PC with a dedicated graphics card
 - D) A server running Linux to host websites

4. How does an Embedded OS differ from a general-purpose OS in terms of size?
- A) Embedded OS is larger than general-purpose OS
 - B) Embedded OS is extremely small (sometimes just kilobytes) while a general-purpose OS requires several gigabytes**
 - C) Both use the same amount of storage space
 - D) Embedded OS is stored in RAM so size does not matter
5. Which of the following is a key characteristic of an Embedded OS?
- A) Very low power consumption, small footprint, and designed for one specific set of tasks**
 - B) Capable of running thousands of different applications
 - C) Requires frequent updates to remain functional
 - D) Designed for large screen displays and keyboard input
6. An ATM machine uses an Embedded OS to manage cash dispensing, receipt printing, and card reading. Why is an Embedded OS appropriate here?
- A) The ATM needs to perform only these specific tasks reliably; it does not need the general-purpose functionality of Windows or macOS**
 - B) Embedded OS makes the ATM machine run faster over a network
 - C) Embedded OS allows the ATM to run any application the bank installs
 - D) Embedded OS is required by government regulations for banking
7. Which of the following statements about an Embedded OS is FALSE?
- A) It is designed for one specific device and set of tasks
 - B) It uses very little memory and power
 - C) It supports downloading and running any application from an app store**
 - D) It is often stored permanently in ROM
8. Which of the following devices does NOT typically contain an Embedded OS?
- A) Digital camera
 - B) Microwave oven
 - C) A student's laptop running Windows 11**
 - D) Smart TV
-

Topic 1.8.3: Network Operating System (NOS)

1. What is a Network Operating System (NOS)?
- A) An OS installed on each individual computer for better internet speed
 - B) An OS designed to manage and support multiple computers connected in a network, enabling them to share resources**
 - C) An OS optimised for downloading files from the internet as quickly as possible
 - D) An OS designed to protect computers from network attacks

2. Which of the following is a primary function of a NOS?
- A) Running video games across multiple computers
 - B) Allowing multiple computers to share files, printers, and internet connections through a central server**
 - C) Increasing the processing speed of individual computers
 - D) Installing applications on individual computers automatically
3. In the school computer lab scenario, what does the NOS central server allow?
- A) Every student to log in to any computer, access the same personal files, and print to the same shared printer**
 - B) Students to play networked video games
 - C) Each computer to run a different operating system
 - D) Teachers to control the content of individual student screens
4. Which of the following is an example of a Network Operating System?
- A) Windows 11
 - B) Android
 - C) Windows Server**
 - D) macOS
5. What is the primary focus of a NOS compared to a general-purpose OS?
- A) A general-purpose OS is focused on security; a NOS is focused on speed
 - B) A NOS focuses on communication, coordination, and resource sharing between multiple connected computers; a general-purpose OS focuses on one user on one machine**
 - C) A NOS can only run on one computer at a time
 - D) A general-purpose OS supports more devices than a NOS
6. In which of the following environments would a NOS be MOST useful?
- A) A single person using a home computer to write documents
 - B) A hospital where patient records must be accessible by authorised staff from any terminal on the network**
 - C) A smartwatch that monitors heart rate
 - D) A digital camera that stores photos
7. What would happen to a school computer lab if the NOS server crashed?
- A) Individual computers continue working normally because they each have their own OS
 - B) Students lose access to shared files, shared printers, and potentially cannot log in — because the server manages centralised authentication and resources**
 - C) All computers restart automatically to reconnect to the server
 - D) Only the teacher's computer is affected
-

Topic 1.8.4: Mobile Operating System

1. What is a Mobile Operating System?
 - A) An OS for running applications on desktop computers connected to mobile networks
 - B) An OS designed specifically for smartphones, tablets, smartwatches, and other handheld devices**
 - C) An OS that allows computers to access mobile data networks
 - D) A lightweight version of Windows designed for older computers
2. Which of the following is the MOST important design priority of a mobile OS that is NOT shared with typical desktop OS?
 - A) Support for a mouse and keyboard
 - B) Aggressive battery efficiency to maximise performance on limited battery power**
 - C) Support for large high-resolution monitors
 - D) The ability to install any software without restriction
3. Android is developed by which company?
 - A) Apple
 - B) Microsoft
 - C) Google (based on Linux)**
 - D) Samsung
4. iOS is used exclusively on which devices?
 - A) Samsung and Xiaomi phones
 - B) Any device that supports touchscreen input
 - C) iPhone and iPad (Apple devices)**
 - D) Huawei devices
5. Approximately what percentage of the global smartphone market does Android hold?
 - A) About 27%
 - B) About 50%
 - C) About 72%**
 - D) About 90%
6. Which mobile operating system is developed by Huawei?
 - A) HarmonyOS**
 - B) Android
 - C) iOS
 - D) Symbian
7. What is "sandboxing" in the context of mobile OS security?
 - A) Each app runs in isolation and cannot access other apps' data without explicit permission — preventing one app from reading another app's private information**

- B) A testing environment where developers test apps before publishing them
 - C) A physical area in a factory where phones are manufactured
 - D) A system that limits how much storage each app can use
8. How does app installation on a mobile OS differ from a desktop OS?
- A) Both mobile and desktop OS require downloading from the manufacturer's website
 - B) Mobile OS uses curated app stores (Google Play, App Store) where apps are reviewed for safety; desktop OS allows downloading from any website**
 - C) Mobile OS does not allow installing any additional apps
 - D) Desktop OS uses app stores exclusively; mobile OS allows downloading from any source
9. Which input method is the primary design focus of a Mobile OS?
- A) Mouse and keyboard
 - B) Touchscreen and voice**
 - C) Gamepad and joystick
 - D) Stylus pen only
10. Which of the following is a feature of mobile OS that is NOT typically built into standard desktop OS?
- A) Support for USB keyboards
 - B) Ability to open multiple windows
 - C) Integration with built-in sensors like accelerometer, gyroscope, GPS, and fingerprint reader**
 - D) Support for external hard drives
11. In which settings on an Android phone can you view and control individual app permissions (camera, microphone, location)?
- A) About Phone → Permissions
 - B) Settings → Apps → [App Name] → Permissions**
 - C) Developer Options → App Permissions
 - D) Settings → Display → App Controls
-

Section B: Short Questions (SQs)

Topic 1.1: Introduction to Operating System

1. Define an operating system. In your answer, mention three functions it performs.
2. Explain why the operating system is described as the "central controller" of a computer system.
3. What was GM-NAA I/O, and why is it historically significant?

4. Using the school principal analogy, explain how an operating system manages two programs that need the same hardware resource at the same time.
 5. Name four examples of operating systems and state which type of device each is used on.
 6. Explain the role of the OS as a translator between the user and the hardware. Use the example of pressing the letter 'A' on a keyboard in your answer.
 7. What is binary language? Why must the OS translate between human input and binary?
 8. Give the binary representation of the letter 'A' as described in this chapter and briefly explain why this representation matters.
 9. List five responsibilities of an operating system in a multi-user environment.
 10. Differentiate between "user isolation" and "resource sharing" as responsibilities of the OS in a multi-user environment.
 11. What is meant by "authentication" in a multi-user OS environment? Give one example of how it is implemented.
 12. Define the three types of user accounts in a modern OS (Administrator, Standard User, Guest) and state one thing each type can and cannot do.
 13. Why does a Standard User account not have the ability to install new software? What is the security benefit of this restriction?
 14. A student logs into the school computer lab and saves an essay. Explain how user accounts ensure that only that student can access their essay.
 15. List the five components that make up a user account in a modern operating system.
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Topic 1.2: Architecture of an Operating System

1. Differentiate between the kernel and the shell. Include what each component does and which layer it occupies.
2. Using the car analogy, explain the relationship between the kernel and the shell.
3. Name the two types of shells and give one example of each.
4. State four specific tasks performed by the kernel.
5. Explain what happens step-by-step when a user types `dir` in the Windows Command Prompt. Trace the command from shell to kernel to hardware.
6. Describe the five layers of OS architecture from Layer 0 (Hardware) to Layer 4 (User Interface), naming each layer and its primary role.
7. Explain why the layered OS design is described as being similar to a skyscraper. What does each "floor" represent?

8. Using the school analogy for OS layers, match each school role (students, administration, guards, library, building) to its corresponding OS layer.
 9. Give two advantages of the modular, layered OS design. Explain each advantage in one to two sentences.
 10. What is a system library? Give two examples of tasks that system libraries handle for application developers.
 11. What is a device driver? Explain why every hardware device needs its own driver.
 12. Draw or describe the relationship between the OS, a device driver, and a hardware device using the translator diagram from the chapter.
 13. Give four examples of device drivers and state what hardware each communicates with.
 14. Explain what would happen if you connected a new printer to your computer but the printer driver was not installed.
 15. Differentiate between a system library and a device driver. State one similarity and two differences.
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Topic 1.3: Process Management

1. Define the term "process" as used in operating system theory.
2. Describe Margaret Hamilton's contribution to the Apollo 11 mission and explain what operating system principle she applied.
3. Name all five states in the process lifecycle and describe what occurs in each state.
4. Draw and label the process state diagram, showing all five states and the transitions between them with their triggers.
5. Explain what happens during the "Creation" phase of a process lifecycle, using the example of opening Google Chrome.
6. What triggers a process to move from the "Running" state to the "Waiting" state? Give one specific example.
7. What happens when a process reaches the "Terminated" state? What does the OS do with the resources the process was using?
8. Define "multitasking" and explain how it is possible for a single CPU core to give the user the impression that multiple programs run simultaneously.
9. Differentiate between "multitasking" and "concurrency." Are they the same concept? Explain.
10. Explain time-slicing. How does it relate to multitasking and concurrency?
11. What is process scheduling? Why is it necessary in an OS?

12. Explain the First-Come, First-Served (FCFS) scheduling method using the analogy of a shop queue.
 13. Using the FCFS method, given P1 (arrives 0s, needs 5s), P2 (arrives 1s, needs 3s), P3 (arrives 2s, needs 2s): state the order in which the CPU processes them and calculate each process's waiting time.
 14. Draw the CPU timeline diagram for the FCFS example in question 43.
 15. Define the "convoy effect" in FCFS scheduling. Give a real-world analogy that illustrates it.
 16. State two advantages and two disadvantages of FCFS scheduling.
 17. Explain why FCFS scheduling is not suitable for real-time operating systems.
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Topic 1.4: Memory

1. Define RAM and explain why it is called "primary memory."
 2. Explain four key characteristics of RAM: speed, volatile nature, size, and purpose.
 3. Using the "desk and filing cabinet" analogy, explain the difference between RAM and a hard drive.
 4. What happened to the IBM 5150 computer's RAM in 1981? How does this compare to modern DDR5 RAM?
 5. List four types of data that RAM holds while a computer is running.
 6. Explain why having more RAM leads to better computer performance.
 7. Define virtual memory and explain when the OS uses it.
 8. Describe the five steps of how virtual memory works when RAM fills up.
 9. What is "paging" or "swapping" in the context of virtual memory? Describe it in two to three sentences.
 10. Using the "desk and backpack" analogy, explain how virtual memory extends the computer's available working memory.
 11. What is "thrashing"? List three symptoms a user might notice when thrashing is occurring.
 12. Compare the data transfer speeds of RAM (DDR5), NVMe SSD, regular SSD, and hard drive. What conclusion does this comparison lead to about virtual memory performance?
 13. Explain the trade-off involved in using virtual memory as an alternative to having more physical RAM.
 14. Differentiate between the Windows "page file" and the Linux/macOS "swap space."
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Topic 1.5: Processes vs. Threads

1. Define the term "thread" as used in operating system theory.
 2. Differentiate between a process and a thread using a clear comparison table covering: definition, memory, independence, resource cost, and an example.
 3. Using the restaurant kitchen analogy, explain the relationship between a process and its threads.
 4. Explain, using the Chrome browser example, how multiple threads work inside one process simultaneously.
 5. Why does a thread crash have a potentially greater impact within its own process than a process crash has on other processes?
 6. Give three examples of tasks that separate threads might handle within a single web browser process.
 7. Explain why threads are described as "cheaper" to create than new processes.
 8. Compare opening 10 browser tabs in one window (threads) versus 10 separate browser windows (processes) in terms of RAM usage and crash isolation.
 9. Define "multithreading" and explain how it differs from simply running multiple processes.
 10. List four benefits of multithreading and provide a brief real-world example illustrating each benefit.
 11. Explain, using the word processor example, how multithreading improves the responsiveness of an application.
 12. Give one example from a video game that illustrates the "Support for Concurrent Operations" benefit of multithreading.
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Topic 1.6: System Calls

1. Define a "system call" and explain its purpose in an operating system.
2. Explain why system calls are necessary. What would happen if programs could directly access hardware without them?
3. Describe the boundary between "user space" and "kernel space" in the system call diagram.
4. Define each of the four system call types: `open()`, `read()`, `write()`, and `fork()`. Give one real-world example for each.
5. A student is typing a report in a word processor and presses Ctrl+S to save. Which system call is primarily used? Describe what happens step by step.

6. When you play a song in a music player, which sequence of system calls is involved from opening the file to reading the audio data? Explain the order.
 7. When a new browser tab opens a new process, which system call is involved? Explain what it does.
 8. State the approximate number of system calls in Linux/macOS and Windows. What does this difference suggest about the complexity of these operating systems?
 9. Explain why system calls improve both the security and the stability of an operating system.
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Topic 1.7: File System Structure and Management

1. Define a "file" and list four different types of data that files can contain.
 2. What four components does every file have? Briefly describe each.
 3. What is a folder (directory)? Explain the type of structure folders create when organising files.
 4. Define the term "file path" and give one example of a complete file path in Windows format.
 5. What is metadata? Explain why it is described as "data about data."
 6. List five specific examples of metadata that a photo file might have.
 7. Explain three ways the OS uses metadata to help users manage their files.
 8. Describe the potential privacy risk of photos containing GPS location metadata when shared online.
 9. Define a "file system" and explain the four main things it manages on a storage device.
 10. Compare FAT32 and NTFS across three criteria: compatibility, maximum file size, and security features.
 11. Why would a user be unable to copy a 5 GB video file onto a USB drive formatted with FAT32?
 12. Which file system would you recommend for a USB drive that needs to work on both Windows and Mac computers without a 4 GB file size limit? Justify your answer.
 13. Explain what "journaling" means in a file system and why it is a valuable feature.
 14. Using the librarian analogy from the chapter, explain how a file system organises data on a storage drive.
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Topic 1.8: Types of Operating Systems

1. Define a Real-Time Operating System (RTOS) and explain why the timing of task completion is critical in RTOS applications.

2. Compare an RTOS with a general-purpose OS (like Windows) using four criteria: response time, optimisation focus, acceptable deadline behaviour, and use cases.
 3. List five real-world applications that use an RTOS and briefly state why real-time processing is essential for each.
 4. Define an Embedded Operating System (EOS). List four key characteristics that distinguish it from a general-purpose OS.
 5. Give five examples of devices that use an Embedded OS and briefly describe what the OS does in each device.
 6. Compare a general-purpose OS and an Embedded OS using a table covering: applications supported, storage size required, update frequency, power usage, and where they are used.
 7. Define a Network Operating System (NOS) and explain its primary purpose.
 8. Describe three specific functions that a NOS performs in a school computer lab environment.
 9. Compare a general-purpose OS and a NOS using four criteria: primary focus, resource sharing capabilities, user management, and typical environment.
 10. Define a Mobile Operating System and list six key design priorities that distinguish it from a desktop OS.
 11. Compare Android and iOS: who created each, what devices they run on, and their approximate market share.
 12. What is "sandboxing" in mobile OS security? Explain why this approach is safer than allowing apps to access any data on the device.
 13. Compare a mobile OS and a desktop OS across six criteria: input method, power management, app installation, app isolation, screen size, and processor type.
 14. Briefly describe the early mobile OS "Symbian" and explain how it differs from modern mobile operating systems like Android and iOS.
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Section C: Long / Extensive Questions (LQs)

Topic 1.1: Introduction to Operating System

LQ1. *Comprehensive Explanation: The Operating System as Central Controller*

- a. Define an operating system. In your answer, identify the three primary functions it performs.
- b. Using the school principal analogy in detail, explain how each of the three OS functions maps to a specific action the principal performs in managing the school.
- c. Trace what happens — step by step — from when a user presses the letter 'A' on a keyboard

to when it appears on screen. Identify what the OS does at each step.

d. Explain why the OS is described as a "universal layer." What would software development look like without this layer?

e. Evaluate: Why is it essential that ALL hardware access goes through the OS rather than directly from applications?

LQ2. Case Study: Multi-User Environment Management

A government office has 50 employees sharing one server. Some employees are department managers (Administrators), most are regular staff (Standard Users), and some desks are available for visiting consultants (Guest accounts).

a. Describe five specific responsibilities the OS must fulfil to manage this environment correctly.

b. Explain what "user isolation" means and describe two scenarios in which its failure would cause a serious problem.

c. Compare the three account types (Administrator, Standard User, Guest) in a table that covers: what they can install, file access, settings control, and persistence of saved data.

d. A consultant (Guest user) accidentally tries to install a software tool for a presentation. Trace exactly what happens: which OS component intervenes, what error occurs, and why this outcome is correct behaviour.

e. Justify why giving every employee Administrator-level access would be dangerous from a security perspective.

Topic 1.2: Architecture of an Operating System

LQ3. Architectural Deep Dive: Kernel, Shell, and OS Layers

a. Define the kernel and the shell. Explain the specific role of each.

b. Using the car analogy (engine vs. steering wheel/dashboard), draw or describe a diagram showing how the user, shell, kernel, and hardware relate to one another. Label all components.

c. Explain the five layers of OS architecture from Layer 0 (Physical Hardware) to Layer 4 (User Interface). For each layer, state: its name, what it contains, and what it does.

d. Using the school analogy (building, guards, administration, library, teachers/students), map each school role to its corresponding OS layer and justify each mapping.

e. Discuss two advantages and two disadvantages of the modular layered OS design. Use specific examples (e.g., driver updates, security boundaries) to support your points.

f. Analyse: If the kernel layer (Layer 2) failed completely, how would this affect each of the other layers? Explain the cascading impact.

LQ4. Comparative Analysis: System Libraries vs. Device Drivers

- a. Define system libraries. Explain the problem they solve for software developers using the "baking a cake" analogy.
 - b. Define device drivers. Explain why each hardware device needs its own driver using the "different electrical languages" concept.
 - c. Construct a comparison table for system libraries and device drivers across the following criteria: definition, purpose, who writes them, which OS layer they belong to, what happens when they are missing, and one example.
 - d. Trace the sequence of events — involving both system libraries and device drivers — that occurs when a music player opens and plays an audio file.
 - e. Evaluate: A developer is writing a new image editing application. At what points in their development will they rely on system libraries? At what points will device drivers become relevant to their application's performance?
-

Topic 1.3: Process Management

LQ5. *Full Process Lifecycle Analysis*

- a. Define "process" and explain how it differs from a program stored on a hard drive.
 - b. Describe all five states in the process lifecycle (New, Ready, Running, Waiting, Terminated). For each state, explain what triggers entry into that state and what triggers the exit.
 - c. Draw the complete process state diagram with all five states, all arrows between them, and labels on each arrow indicating the trigger for that transition.
 - d. Trace the complete lifecycle of Google Chrome from the moment the user double-clicks the icon to the moment they click the close button (X). Identify which state Chrome is in at each stage, including at least one transition through the "Waiting" state.
 - e. Explain how Margaret Hamilton's work on the Apollo 11 computer OS applied the principles of process management to a life-or-death situation.
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LQ6. *Multitasking, Concurrency, and FCFS Scheduling*

- a. Define multitasking and concurrency. Explain the distinction between them with a specific example involving three programs.
- b. Explain time-slicing. Include a diagram (timeline) showing how three processes share one CPU over a period of time using time-slicing.
- c. Define FCFS scheduling. State the single rule that governs FCFS.

Given the following processes:

Process	Arrival Time	Burst Time (Time Needed)
P1	0s	4s
P2	1s	6s
P3	3s	2s
P4	5s	3s

d. Apply FCFS scheduling to the table above:

- Draw the CPU execution timeline
- Calculate the Start Time and Finish Time for each process
- Calculate the Waiting Time for each process
- Calculate the Average Waiting Time

e. Discuss two advantages and two disadvantages of FCFS scheduling. Use your calculated results from part (d) to illustrate the convoy effect disadvantage with specific numbers.

f. Evaluate: Is FCFS a fair scheduling algorithm? What does "fair" mean in the context of scheduling? Who benefits and who suffers under FCFS?

Topic 1.4: Memory

LQ7. RAM and Virtual Memory: A Complete Analysis

a. Define RAM. Explain the four key characteristics (speed, volatility, size, purpose) with a specific example of what RAM holds while you work on a school project.

b. Using the "desk and filing cabinet" analogy, construct an extended explanation that covers:

- What the desk represents and why its size matters
- What the filing cabinet represents
- What happens when you run out of desk space (connecting directly to virtual memory)

c. Define virtual memory. Trace the five steps of how virtual memory works when RAM becomes full, using an example of a student running a video editing project that requires 6 GB on a computer with only 4 GB of RAM.

d. Construct a comparison table for RAM (DDR5), NVMe SSD, Regular SSD, and Hard Drive covering: approximate speed, role in the memory system, and cost implications.

e. Define "thrashing." Explain the user-visible symptoms and the underlying OS behaviour that causes it.

f. Analyse the following scenario: A student's computer has 4 GB of RAM. They have open: a web browser (2 GB), a music player (200 MB), a word processor (300 MB), and they want to launch a game requiring 3 GB.

- Can they open the game? Show your calculation.
- What will the OS do if they try anyway?
- What are two solutions to this problem?

g. Evaluate: Is virtual memory a good long-term solution for insufficient RAM, or should users always install more physical RAM? Justify your answer using the speed comparison data.

Topic 1.5: Processes vs. Threads

LQ8. Processes, Threads, and Multithreading

- Define "process" and "thread." Clearly explain the core difference between them in terms of memory, independence, and resource cost.
 - Using the restaurant kitchen analogy (kitchen = process, cooks = threads), explain:
 - What happens to the cooks if the kitchen closes (process terminates)
 - What happens to the other cooks if one cook takes a break (thread pauses)
 - Why sharing the same kitchen (memory space) helps the cooks work faster together
 - Construct a comparison table for processes and threads across six criteria: definition, memory space, independence, resource cost, communication, and one example.
 - Using the Chrome browser example with three threads (loading webpage, playing video, downloading file), explain:
 - What these three tasks share (and why that is efficient)
 - What would happen if each tab were a separate process instead
 - Which approach uses less RAM and why
 - Define multithreading. Explain all four benefits of multithreading with a specific real-world example for each benefit.
 - Design a multithreaded application: You are designing a social media app for smartphones. Identify at least five tasks this app needs to perform simultaneously and assign each to a thread. Justify why each task benefits from being a separate thread rather than running sequentially.
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Topic 1.6: System Calls

LQ9. System Calls: The Gateway to the Kernel

- Define a system call. Explain the purpose it serves and why it exists.
- Draw and describe the system call boundary diagram showing: user space, the system call interface, kernel space, and hardware. Label what happens at each level when a program saves a file.

- c. Define and explain all four types of system calls (`open`, `read`, `write`, `fork`). For each:
- State its name and definition
 - Describe what it does technically
 - Give a real-world example
- d. Trace the complete sequence of system calls involved in the following scenario: A student uses a photo editing app to open a photo, applies a filter, and saves the result as a new file. Identify every system call that occurs in order.
- e. Analyse: Why do security researchers care deeply about how system calls are implemented? What could go wrong if a flaw existed in the `fork()` or `write()` system call implementation?
- f. Evaluate the statement: *"System calls make programs slower because they force every hardware action to go through the kernel."* Do you agree or disagree? Justify your position, considering both the performance cost and the security/stability benefit.
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Topic 1.7: File System Structure and Management

LQ10. File Systems: Organising the Digital World

- a. Define files, folders, and metadata. For each, give two examples drawn from a student's typical computer usage.
- b. Draw the Windows-style file system hierarchy tree that includes the following: a root drive (C:), a Users folder, two user sub-folders (Ahmed and Sara), each with Documents, Pictures, and Music folders, each containing at least two files.
- c. Given the hierarchy you drew in part (b), write the complete file path for:
- Ahmed's homework document
 - Sara's essay
 - A music file in Ahmed's Music folder
- d. Construct a metadata table for a photo file, listing at least eight different metadata fields and realistic example values for each.
- e. Construct a comparison table for five file systems (FAT32, NTFS, APFS/HFS+, EXT4, exFAT) covering: operating system used on, maximum file size, key features, and one limitation.
- f. Analyse this scenario: A videographer wants to copy a 10 GB raw video file from their Windows laptop to a USB drive to give to a colleague who uses a Mac.
- Why would FAT32 fail for this task?
 - Why would NTFS be problematic for the Mac user?
 - Which file system should the USB drive use? Justify your recommendation.
- g. Evaluate: What would the experience of using a computer be like if file systems did not exist? Describe at least three specific problems that would arise.

Topic 1.8: Types of Operating Systems

LQ11. OS Types: Matching the System to the Task

- a. Name and define all four types of operating systems covered in this chapter: RTOS, EOS, NOS, and Mobile OS.
- b. Construct a comprehensive comparison table for all four OS types across these criteria:
- Full name and acronym
 - Primary purpose
 - Key defining characteristic
 - Two examples of devices or environments where it is used
 - One feature it prioritises above others
 - One feature it does NOT need that a general-purpose OS has
- c. Explain why an RTOS is essential for an anti-lock braking system (ABS) in a car. Include in your explanation:
- What an RTOS guarantees that a general-purpose OS does not
 - The specific consequence of a deadline miss in this scenario
 - Why Windows or Android would be inappropriate choices
- d. An engineering company wants to install an operating system in each of the following: (1) a smartwatch, (2) a hospital patient monitoring system, (3) a school computer lab network, (4) a coffee machine. For each device, recommend the most appropriate type of OS (RTOS, EOS, NOS, or Mobile OS) and justify your recommendation with two specific reasons.
- e. Analyse the evolution of mobile operating systems from early Symbian phones to modern Android and iOS. Discuss three major capabilities that modern mobile OS added that early mobile OS could not support.
- f. Evaluate: Could a single, universal operating system be designed that works equally well for all four environments (real-time, embedded, network, and mobile)? Discuss the trade-offs involved and explain why specialised OS types exist.

LQ12. System Design: End-to-End Request Processing

A student is at a school computer using a NOS-connected computer. They open a browser, type a search query, and receive results from a web server.

- a. Identify which type of OS is running on: (1) the student's individual computer, (2) the school network server, (3) a web server handling the search query.
- b. Trace the complete journey of the student's request from the moment they press Enter, covering:
- What the shell does

- What system calls are involved
 - How the OS manages the browser process
 - How the NOS routes the request across the school network
 - How the web server's OS handles the incoming query
- c. At each stage of the request processing, identify which OS component (shell, kernel, device driver, system library, process, thread, file system, system call) is involved and explain its specific role.
- d. Explain how each of the following OS features contributes to making this scenario work smoothly and securely: multitasking, system calls, user accounts, and network OS resource sharing.
- e. Discuss what would break in this scenario if: (1) the network device driver was corrupt, (2) the browser process crashed mid-request, (3) the NOS server went offline.
- f. Evaluate: The student notices the search results load slowly. Using your knowledge of RAM, virtual memory, process scheduling, and multithreading, suggest three technical reasons why slowness might occur and one remedy for each.

End of Comprehensive Question Bank: Unit 1 — Operating Systems: Structure and Services

Question Count Summary:

Section	Total Questions
Section A: MCQs	226
Section B: Short Questions	110
Section C: Long Questions	12 (multi-part)
Total	348 items